

Screen Dimensions

$$W = 155 \text{ mm}$$

$$L = 107 \text{ mm}$$

Rotations to Traverse

$$W = 4(2\pi) + \frac{\pi}{2} = \frac{17\pi}{2} \text{ rad}$$

$$L = 3(2\pi) = 6\pi \text{ rad}$$

Rate of Traverse

$$W = \frac{155 \text{ mm}}{\frac{17\pi}{2} \text{ rad}} = 5.8045 \text{ mm/rad}$$

$$L = \frac{107 \text{ mm}}{6\pi \text{ rad}} = 5.6765 \text{ mm/rad}$$

External Dimensions

$$W = 9''$$

$$L = 7.26''$$

$$H = 1.125''$$

Bottom Track

$$W = 221 \text{ mm}$$

$$L = 180 \text{ mm}$$

$$H = 3 \text{ mm}$$

Note: Etch A Sketch operates on a cable-driven, sliding rail system. The cables are somewhat slippery, so the knob will continue to spin if continually driven, even if the stylus is at its limit.

→ This mechanic can be leveraged to calibrate a stepper motor system without the use of limit switches.

→ Just drive the stepper the full screen dimension back to the origin, and the cursor will settle at the origin, while the knob keeps spinning without harm to the mechanism.

→ Starting position may be arbitrary.